

Task 1:

Dividing 192.168.10.0/24 into 4 Equal Subnets

Objective: Subnet the base 192.168.10.0/24 network into four equal subnets and specify the CIDR prefix, subnet mask, network address, broadcast address, and host ranges for each.

1. Subnetting Parameters Calculation:

- **Base Network:** 192.168.10.0
 - **Original Mask:** /24 (255.255.255.0)
 - **Subnets Required:** 4
 - **Borrowed Network Bits (n):** $2^n = 4 \Rightarrow n = 2$ bits
 - **New Prefix Length:** $24 + 2 = /26$
 - **New Subnet Mask:** 255.255.255.192
 - **Block Size (Increment Step):** $256 - 192 = 64$
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2. 4 Equal Subnets Details Table:

Su bn et #	CIDR Notati on	Subn et Mask	Netw ork Addr ess	First Usab le Host IP	Last Usab le Host IP	Broa dcas t Addr ess
Su bn et 1	192.16 8.10.0 /26	255.2 55.25 5.192	192.1 68.10 .0	192.1 68.10 .1	192.1 68.10 .62	192.1 68.10 .63
Su bn et 2	192.16 8.10.6 4/26	255.2 55.25 5.192	192.1 68.10 .64	192.1 68.10 .65	192.1 68.10 .126	192.1 68.10 .127
Su bn et 3	192.16 8.10.1 28/26	255.2 55.25 5.192	192.1 68.10 .128	192.1 68.10 .129	192.1 68.10 .190	192.1 68.10 .191
Su bn et 4	192.16 8.10.1 92/26	255.2 55.25 5.192	192.1 68.10 .192	192.1 68.10 .193	192.1 68.10 .254	192.1 68.10 .255

Task 2:

Device IP Assignment for Subnet 1 (192.168.10.0/26)

Objective: Assign valid host IP addresses from Subnet 1 (192.168.10.0/26) to 5 network devices while excluding the reserved Network and Broadcast addresses.

1. Subnet 1 Boundaries:

- **Subnet Network Address:** 192.168.10.0 (*Reserved - Unusable for hosts*)
 - **Subnet Broadcast Address:** 192.168.10.63 (*Reserved - Unusable for hosts*)
 - **Subnet Mask:** 255.255.255.192
 - **Valid Usable Host IP Range:** 192.168.10.1 to 192.168.10.62
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2. Device IP Address Allocation Table:

Device Name	Assigned IP Address	Subnet Mask	Default Gateway	Allocation Type & Purpose
Default Gateway Router	192.168.10.1	255.255.255.192	N/A (Gateway Node)	Static: Assigned to router's LAN interface (First usable IP).
Work Laptop 1	192.168.10.10	255.255.255.192	192.168.10.1	DHCP / Static: Workstation connected via Ethernet/Wi-Fi.

Device Name	Assigned IP Address	Subnet Mask	Default Gateway	Allocation Type & Purpose
Personal Laptop 2	192.168.10.11	255.255.255.192	192.168.10.1	DHCP / Static: Secondary development machine.
Network Printer	192.168.10.20	255.255.255.192	192.168.10.1	Static: Dedicated static IP for shared network printing.

Device Name	Assigned IP Address	Subnet Mask	Default Gateway	Allocation Type & Purpose
Mobile Smartphone	192.168.10.30	255.255.255.192	192.168.10.1	DHCP: Wireless smartphone host on local WLAN.

3. Compliance Rules Applied:

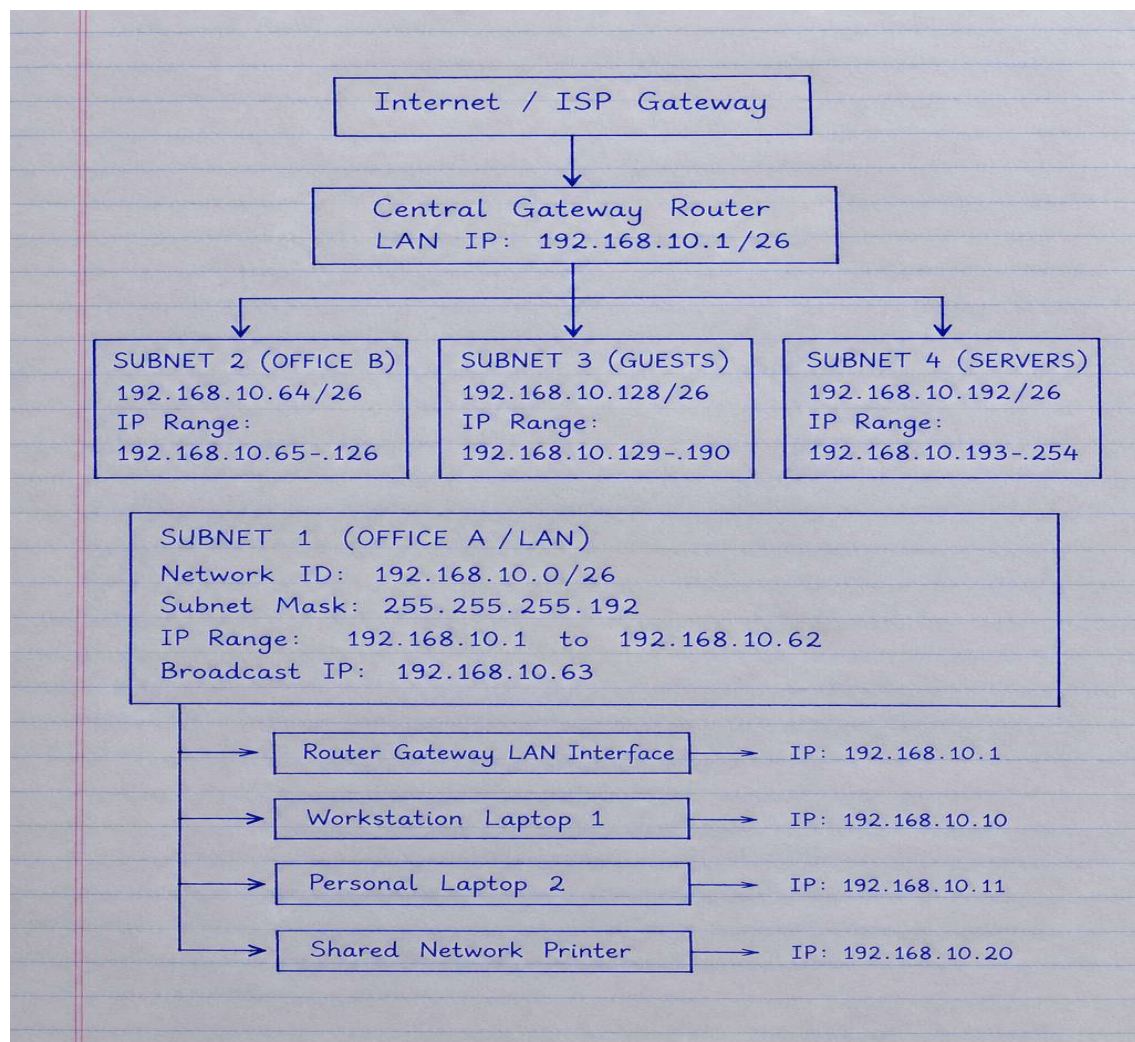
- 1. Reserved Address Protection:** Neither 192.168.10.0 (Network ID) nor 192.168.10.63 (Broadcast ID) were assigned to any physical host.
- 2. Subnet Scope:** All assigned IPs lie strictly within the valid range of 192.168.10.1 – 192.168.10.62.
- 3. Gateway Uniformity:** All end-host devices point to 192.168.10.1 (Router) as their default exit point to external networks.

Task 3:

Subnet Plan Network Topology Diagram

Objective: Design a visual network topology diagram illustrating the 4 created subnets (192.168.10.0/26 to 192.168.10.192/26), including IP ranges, gateway interface connections, and device endpoint mapping for Subnet 1.

1. Visual Network Topology Diagram:



2. Network Plan & Device Connectivity Summary:

- **Central Router:** Interconnects all four subnets via internal routing interfaces.
- **Subnet 1 Mapping:**
 - **Network Address:** 192.168.10.0/26
 - **Usable IP Range:** 192.168.10.1 – 192.168.10.62
 - **Connected Devices:** Router Interface (.1), Work Laptop 1 (.10), Personal Laptop 2 (.11), Network Printer (.20), and Smartphone (.30).
- **Subnets 2, 3, and 4 Mapping:**
 - **Subnet 2:** Reserved for Office B (192.168.10.64/26, Range: .65 – .126).
 - **Subnet 3:** Reserved for Guest Wi-Fi (192.168.10.128/26, Range: .129 – .190).
 - **Subnet 4:** Reserved for Server Infrastructure (192.168.10.192/26, Range: .193 – .254).

Task 4:

Step-by-Step Subnetting Explanation (Beginner-Friendly Guide)

Objective: Explain the mathematical logic and step-by-step procedure used to calculate subnet masks, valid host counts, block sizes, and IP assignments in simple, conversational language.

Step 1: Understanding Our Base Starting Point

Hey bro! Imagine we are given one big network block:

192.168.10.0/24.

- The **/24** part means the first 24 bits belong to the network highway, leaving **8 bits** for device IP addresses ($32 - 24 = 8$ bits).
 - That gives us $2^8 = 256$ total IP addresses in total (192.168.10.0 to 192.168.10.255).
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Step 2: Calculating the New Subnet Mask for 4 Subnets

We need to divide this big network into **4 equal subnets**.

1. To find out how many bits to borrow, we use the binary formula: $2^n = \text{Number of Subnets}$.

2. Here, $2^2 = 4$, so we need to **borrow 2 bits** from the host section.
3. We add these 2 borrowed bits to our original /24 prefix length: $24 + 2 = /26$.
4. **Calculating Mask Bytes:** In binary, a /26 mask has 26 ones followed by 6 zeros:

11111111.11111111.11111111.11000000

5. Converting 11000000 to decimal gives $128 + 64 = 192$.

- **Our New Subnet Mask: 255.255.255.192**

Step 3: Calculating Total IPs & Usable Hosts per Subnet

After borrowing 2 bits, we have **6 host bits remaining** ($32 - 26 = 6$).

1. **Total IPs per Subnet:** $2^6 = 64$ IP addresses.
2. **The 2 Reserved IPs Rule:** In networking, you can never assign the very first IP (Network ID) or the very last IP (Broadcast ID) to an individual device.
3. **Usable Hosts Calculation:** $64 - 2 = 62$ usable host IP addresses per subnet.

Step 4: Finding the Block Size (How IP Ranges Jump)

To quickly list down where each subnet starts and ends, we calculate the **Block Size**:

$$\text{Block Size} = 256 - 192 = 64$$

So each subnet increments by 64:

- **Subnet 1:** 192.168.10.0 to 192.168.10.63
 - **Subnet 2:** 192.168.10.64 to 192.168.10.127
 - **Subnet 3:** 192.168.10.128 to 192.168.10.191
 - **Subnet 4:** 192.168.10.192 to 192.168.10.255
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Step 5: Assigning IPs to Devices (Practical Logic)

Taking **Subnet 1 (192.168.10.0/26)** with usable IPs from **.1 to .62**:

1. **Router Interface (192.168.10.1):** Always give the very first usable IP to the default gateway router so all devices know where to send internet traffic.
2. **Shared Printer (192.168.10.20):** Give static IPs to shared infrastructure like printers so their address never changes when rebooted.
3. **Laptops & Mobile (.10, .11, .30):** Assign remaining middle host IPs dynamically or statically to end-user workstations and smartphones.

